

Recovering the unwritten recurrences of the table OEIS A253985

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2 September 2026

Abstract

OEIS A253985 is a two-dimensional table $T(n, k)$ whose empirical recurrences are, for several of its lines, given only as an ORDER — “k=2: [order 41]” and the like — with the coefficients in a linked file rather than in the entry. Those conjectures can still be settled, and the recurrences recovered. A line of the table is a fixed-width or fixed-height array count, hence a walk count on finitely many vertices, hence satisfies some monic recurrence of order at most that number; Berlekamp–Massey on twice as many exact terms returns the MINIMAL one. Where its order equals the order the entry states — and where the same holds after the first few terms are dropped — any recurrence of that order the line satisfies has a characteristic polynomial that is a multiple of the minimal one and of equal degree, hence is that polynomial. This note settles column 2, column 3.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 The table and the unwritten conjectures

OEIS A253985 is “ $T(n, k)$ = Number of $(n+2) \times (k+2)$ 0..1 arrays with every 3×3 subblock diagonal maximum plus antidiagonal maximum nondecreasing horizontally and vertically.”. It is read by antidiagonals and begins

512, 3372, 3372, 21304, 36434, 21304, 136868, 417518, ...

The entry carries, among its empirical claims:

k=2: [order 41]

k=3: [order 68]

As of the “Last modified” line on the live entry (Jul 23 2025 14:23:35, revision 6) these are still recorded as empirical, and the recurrences themselves are not in the entry text.

2 Why a line of the table is a walk count

Fix the index that the line fixes. The entry defines $T(n, k)$ as a number of arrays of a shape determined by n and k , subject to a condition local to a bounded window of consecutive lines. Substituting the fixed index into the entry’s own wording turns the definition into an ordinary one-parameter array count, and for such a count the admissible windows are the vertices of a finite digraph and an array is a walk. So the line is a walk count on S vertices, and by the Cayley–Hamilton theorem it satisfies a monic integer recurrence of order at most S .

References

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